

LAB 1:

The screenshot shows a SQL IDE interface. At the top, there are two tabs: "Query" and "Query History". The "Query" tab is active, displaying a SQL query with line numbers 1 through 4. The query is: `SELECT rental_id, customer_id FROM rental WHERE return_date IS NULL LIMIT 1;`. Below the query editor, there are three tabs: "Data Output", "Messages", and "Notifications". The "Data Output" tab is active, showing a table with two columns: "rental_id" (type [PK] integer) and "customer_id" (type smallint). The table contains one row with values 11496 and 155 respectively. Above the table, there is a toolbar with various icons for file operations, execution, and formatting.

```
1 SELECT rental_id, customer_id
2 FROM rental
3 WHERE return_date IS NULL
4 LIMIT 1;
```

	rental_id [PK] integer	customer_id smallint
1	11496	155

Giả sử kết quả là:

rental_id = 2

customer_id = 459

Mở transaction và cập nhật trả đĩa

Query	Query History
1	INSERT INTO payment (
2	customer_id,
3	staff_id,
4	rental_id,
5	amount,
6	payment_date
7	)
8	VALUES (
9	459,
10	1,
11	2,
12	99.99,
13	'WRONG_DATE_FORMAT'
14	);

Data Output	Messages	Notifications
ERROR: current transaction is aborted, commands ignored until end of transaction block		
SQL state: 25P02		

Rollback

```
1 BEGIN;  
2  
3 UPDATE rental  
4 SET return_date = NOW()  
5 WHERE rental_id = 2;
```

Data Output Messages Notifications

UPDATE 1

Query returned successfully in 52 msec.

```
1 SELECT rental_id, return_date
2 FROM rental
3 WHERE rental_id = 2;
```

Data Output Messages Notifications

	rental_id [PK] integer	return_date timestamp without time zone
1	2	2026-09-18 14:52:50.149647

Tất cả thành công → COMMIT

Có một bước lỗi → ROLLBACK toàn bộ

Ex1:

Query

Query History

1

BEGIN;

2

3

-- Bước 1: Đánh dấu đã trả DVD

4

UPDATE rental

5

SET return_date = NOW()

6

WHERE rental_id = 2;

7

8

-- Bước 2: Cố tình chèn ngày sai

9

INSERT INTO payment

10

(customer_id, staff_id, rental_id, amount, payment_date)

11

VALUES

12

(459, 1, 2, 99.99, 'WRONG_DATE_FORMAT');

13

14

COMMIT;

Data Output

Messages

Notifications

ERROR: current transaction is aborted, commands ignored until end of transaction block

SQL state: 25P02

Query

Query History

1

SELECT return_date

2

FROM rental

3

WHERE rental_id = 2;

Data Output

Messages

Notifications

ERROR: current transaction is aborted, commands ignored until end of transaction block

SQL state: 25P02

Query

Query History

1

BEGIN;

2

3

UPDATE rental

4

SET return_date = NOW()

5

WHERE rental_id = 2;

6

7

INSERT INTO payment

8

(customer_id, staff_id, rental_id, amount, payment_date)

9

VALUES

10

(459, 1, 2, 99.99, NOW());

11

12

COMMIT;

Data Output

Messages

Notifications

COMMIT

Query returned successfully in 38 msec.

Ex3

Query Query History

1 **ROLLBACK TO SAVEPOINT** staff_created;

Data Output Messages Notifications

ERROR: ROLLBACK TO SAVEPOINT can only be used in transaction blocks

SQL state: 25P01